"PAPER_{0:D3}" -f [int]# PAPER #84 ■ Information Paradox Resolution in 26D UQFF

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Hawking (1976): if T*H is exactly thermal*, *S*BH decreases from S=A/4 to S=0 ? pureness is lost. Unitarity requires S_entanglement (Hawking radiation with BH) to follow the Page curve: first increasing (early time), then decreasing back to 0 (late time).

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Channel Structure

The UQFF 26-layer compressed gravity framework assigns each dimensional layer a quantum channel capacity:

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Total information: `I_{\rm total} = \sum_{k=1}^{26} I_k = S_{\rm BH,initial}`

Channel Assignment During Evaporation

Layer Group	Channels	Storage Type	4D Observable
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5■18	UQFF extended	Sub-Planckian modes	Not observable
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Channels 25■26 host the **complete pre-collapse pure state** throughout evaporation, ensuring global unitarity while channels 1■4 produce the observed thermal spectrum.

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The UQFF Page curve modifies the standard quantum extremal surface (QES) prediction:

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t_P^{\rm UQFF} = t_P^{\rm GR} \times e^{+\kappa \cdot t_{\rm evap}} \approx t_P^{\rm GR}
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Aspect	Standard QM	UQFF 26D	Resolution
Information storage	In BH?	Channels 25■26	Non-local, always preserved
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UQFF Discovery: Novel application of UQFF calibration constants (? = 5.0■10?4 day?■, [SSq] = 0.57) uniquely enabling this analysis ■ establishing a new connection in the UQFF framework not present in Standard Model treatments.

1. The Standard Paradox

Hawking (1976): if T*H is exactly thermal, S*BH decreases from S=A/4 to S=0 ? pureness is lost. Unitarity requires S_entanglement (Hawking radiation with BH) to follow the Page curve: first increasing (early time), then decreasing back to 0 (late time).

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2. 26D Holographic Channel Resolution

Channel Structure

The UQFF 26-layer compressed gravity framework assigns each dimensional layer a quantum channel capacity:

```
I_k = -\text{Tr}[\rho_k \log \rho_k] \quad k = 1, \ldots, 26
```

Total information: `I_{\rm total} = \sum_{k=1}^{26} I_k = S_{\rm BH,initial}`

Channel Assignment During Evaporation

Layer Group	Channels	Storage Type	4D Observable
1■4	UQFF observable	Hawking photons	Thermal spectrum
5■18	UQFF extended	Sub-Planckian modes	Not observable
19■24	UQFF deep structure	Non-local entanglement	Nil
25■26	**Cosmic Egg layers**	**Non-local pure state**	**Information preserved**

Channels 25■26 host the **complete pre-collapse pure state** throughout evaporation, ensuring global unitarity while channels 1■4 produce the observed thermal spectrum.

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3. UQFF Page Curve

The UQFF Page curve modifies the standard quantum extremal surface (QES) prediction:

```
S_{\rm UQFF}(t) = \min\left[S_{\rm thermal}(t), \; S_{\rm BH}(t) + \sum_{k=25}^{26} I_k
(1 - e^{-\kappa t})\right]
```

Where:

- S_thermal(t) = early-time entropy of Hawking radiation
- S_BH(t) = Bekenstein-Hawking entropy (decreasing)
- The $e^{-?t}$ term comes directly from the UQFF ? = 0.0005/day decay
- Page time t*P occurs when S*thermal = S*BH + ?I*channels25-26

UQFF Page Time

```
t_P^{\rm UQFF} = t_P^{\rm GR} \times e^{+\kappa \cdot t_{\rm evap}} \approx t_P^{\rm GR}
\times (1 + \kappa \cdot t_{\rm evap})
```

For stellar BH (t_evap ~ 1074 s): correction factor is astronomically large ■ physically this means that within any finite observation time, the UQFF Page curve looks **thermal** to a 4D observer, with the information recovery pushed beyond any accessible time. This is consistent with the absence of observational evidence for information recovery in Hawking radiation.

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4. Firewall Prevention

The AMPS firewall paradox requires choosing between a smooth horizon (infalling observer) or pure Hawking radiation. The UQFF resolves this through **channels 19■24 non-local entanglement**: the infalling observer encounters a smoothly modified vacuum (Superconductive mode, E_react ■ 10?■■) rather than a firewall, while the external observer's Hawking radiation becomes approximately (not exactly) thermal.

Summary

Aspect	Standard QM	UQFF 26D	Resolution
Information storage	In BH?	Channels 25■26	Non-local, always preserved
Page curve	QES prediction	UQFF ?-modified	Extended beyond t_evap
Firewall	AMPS paradox	SCm smooth horizon	Channels 19■24 entanglement
4D observer	Thermal Hawking	Approximately thermal	Consistent

Source: MAIN1CoAnQi.cpp Batch 21 (InformationParadoxModule) | ? = 0.0005/day | [SSq] = 0.57 .Groups[1].Value "PAPER$_{0:D3}$" -f $n

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UQFF computed: Canonical UQFF buoyancy parameter U*bi* = *?■[SSq]■GM/r■ = 5.0e-4■0.57■6.67e-11■M/r■; for solar parameters: U*bi,Sun = 5.7e-4■6.67e-11■1.99e30/(6.96e8)■ = 1.47e+2 m/s■.